

Research Paper

Prevalence and Multidrug Resistance Pattern of *Salmonella* Carrying Extended-Spectrum β -Lactamase in Frozen Chicken Meat in Bangladesh

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ABSTRACT

Salmonella is an important foodborne pathogen that causes public health problems globally, and the increase of antimicrobial resistance in *Salmonella* has intensified the problem. Chicken meat is an important reservoir and disseminator of *Salmonella* to humans. This study aimed at estimating the burden of *Salmonella* carrying extended-spectrum β -lactamase (ESBL) and their antimicrobial resistance pattern in 113 domestic frozen chicken meat samples purchased from supershops available in five divisional megacities of Bangladesh. The study also focused on the determination of β -lactamase-, and plasmid-mediated quinolone resistance-encoding genes. All samples were analyzed for the presence of *Salmonella* using selective media and PCR assay. Antimicrobial susceptibility test was done by disk diffusion test, and ESBL screening was performed by double-disk synergy tests. Resistance genes were detected using multiplex PCR. Of samples, 65.5% were positive for *Salmonella* spp., and, of these, 58.1% isolates were ESBL producers. All the isolates were multidrug resistant (MDR): 40.5% were resistant to both three to five and six to eight antimicrobial classes; 17.6% were resistant to 9 to 11 classes, and 1.4% isolates to 12 to 15 classes. The highest rates of resistance were observed against oxytetracycline (100%), followed by trimethoprim-sulfamethoxazole (89.2%), tetracycline (86.5%), nalidixic acid (83.8%), amoxicillin (74.3%), and pefloxacin (70.3%). Notably, 48.6% of isolates demonstrated resistance to imipenem. One (1.4%) isolate was possibly extensively drug resistant. All the isolates were positive for the *bla*_{TEM} gene, 2.7% were positive for *bla*_{CTX-M-1}, and 20.3% for *bla*_{NDM-1}. The prevalence of *qnrA* and *qnrS* genes was 4.1 and 6.8%, respectively. This study shows that ESBL-producing *Salmonella* are widespread in frozen chicken meat in Bangladesh, which puts greater responsibility on food processors and policy makers to ensure food safety.

HIGHLIGHTS

- *Salmonella* prevalence was 65.5% in frozen chicken meat; 58.1% of *Salmonella* isolates were ESBL producers.
- All isolates were MDR, and one isolate was possibly extensively drug resistant.
- Of isolates, 68.9% were resistant to 10 or more antimicrobial agents.
- The highest resistance rate was found to oxytetracycline and trimethoprim-sulfamethoxazole.
- Resistance genes such as *bla*_{CTX-M-1}, *bla*_{NDM-1}, *qnrA*, and *qnrS* were identified.

Key words: Antimicrobial resistance; *bla*_{NDM-1}; Extended-spectrum β -lactamase; Frozen chicken meat; Prevalence; *Salmonella*

Foodborne diseases constitute an important public health problem all over the world. According to the World Health Organization, over 600 million people were affected with 32 foodborne diseases caused by bacteria, viruses, parasites, and chemicals; and 420,000 related deaths occurred globally in 2010 (69). Diarrhea-causing pathogens were the leading cause of foodborne diseases, which were responsible for 230,000 deaths globally in 2010 (69).

Among these, *Salmonella* spp. constitute a remarkable public health hazard worldwide (59). It is estimated that nontyphoidal *Salmonella* pathogens account for 93.8 million cases of human gastroenteritis and 155,000 deaths each year globally (41). Chicken is one of the most widely consumed meats in the world; however, it is also recognized as an important reservoir and disseminator of *Salmonella* (33). *Salmonella* is found in the gastrointestinal tract of chickens and often contaminates carcasses during slaughter and processing; therefore, it can be transmitted to humans directly through contact with chickens or indirectly by

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consumption of contaminated chicken meat and meat products (21, 26, 65). Moreover, the emergence of antimicrobial resistance (AMR) in *Salmonella* has led to an increased human health hazard because of increased mortality of infected patients (12, 64, 76). Many *Salmonella* species isolated from chickens and chicken meat showed resistance to several antibiotics (55, 60, 63). Overuse or abuse of antimicrobials in poultry production is an important factor contributing to the emergence, selection, and spread of AMR in *Salmonella* among the poultry population (25). Antimicrobials are extensively used in poultry farming with the aim of treating and preventing poultry diseases as well as improving growth performance; this potentially results in multidrug-resistant (MDR) *Salmonella* (45). The emergence of MDR *Salmonella* has been a growing public health concern throughout the world during the past decades (24, 68, 75). MDR *Salmonella* isolates from chicken meat were found to carry β -lactam, aminoglycosides, tetracycline, and trimethoprim-sulfamethoxazole AMR genes, posing a significant public health threat (24, 63). Furthermore, the emergence and dissemination of *Salmonella* resistant to fluoroquinolones and third-generation cephalosporins limits treatment with the available antibiotics such as carbapenems (e.g., imipenem, meropenem, and ertapenem) (20). It is well established that hydrolytic enzymes such as extended-spectrum β -lactamases (ESBLs) produced by several gram-negative bacteria confer resistance to third- and fourth-generation cephalosporins and monobactams (aztreonam) (10, 48).

ESBL-producing bacteria may also carry the genetic determinants for resistance to aminoglycosides, fluoroquinolones, chloramphenicol, and tetracycline, which is a disconcerting risk to public health (15, 48, 58). ESBL-producing *Salmonella* pathogens are widespread and commonly act as donors for antibiotic resistance transmission via transferable elements, including plasmids carrying antibiotic resistance genes, integrons, and transposons (22, 46). Various resistance genes, such as *bla*_{TEM}, *bla*_{SHV}, and *bla*_{CTX-M}, have been reported to encode broad-spectrum β -lactamase (BSBL) and ESBL activity and to confer β -lactam resistance in *Salmonella* (9, 24, 42, 48, 56). In addition, carbapenemase-encoding genes, mainly *bla*_{KPC}, *bla*_{NDM}, *bla*_{VIM}, and *bla*_{OXA-48}, have been reported; these are responsible for carbapenem resistance in *Enterobacteriaceae*, including *Salmonella* (20, 53). Resistance to fluoroquinolones is mainly caused by plasmid-mediated quinolone resistance (PMQR) genes (*qnrA*, *qnrB*, *qnrC*, *qnrD*, and *qnrS*), and these genes have been identified in *Enterobacteriaceae* in different countries (54). Furthermore, mutations in DNA gyrase (encoded by *gyrA* and *gyrB*) and topoisomerase IV (encoded by *parC* and *parE*) are associated with fluoroquinolone resistance in most of the bacterial species (39, 72).

Raw or undercooked chicken meat has the highest contamination with ESBL-producing *Salmonella* compared with other meat sources (8). The level of *Salmonella* contamination of chicken meat depends mostly on the process of slaughtering, sanitation during processing and packaging, maintenance of adequate cold chain storage from processing to the retail level and the consumer, and

finally, personal hygiene during handling at the retail end (26). Research has been conducted in Bangladesh on *Salmonella* contamination and AMR patterns in raw chicken meat from live bird markets (6, 7, 18, 19, 31, 34, 51). Nowadays, consumers in Bangladesh, especially city dwellers, prefer to go to supershops to buy their daily commodities, including chicken meat, due to easy availability. The supershops sell frozen chicken meat products, and consumers prefer these because they require minimal processing for cooking and are time-saving (32). However, the contamination of this frozen chicken meat with foodborne pathogens has raised the issue of microbiological safety and public health hazards in Bangladesh. Two previous in-county studies reported bacterial contamination in frozen chicken meat in Dhaka city (5, 66). Uddin et al. (66) confirmed the contamination of frozen chicken meat with enterotoxigenic *E. coli*, nonvirulent *Salmonella* spp., and *Shigella* spp. Another study also observed *Salmonella* contamination, along with other organisms, in preprocessed raw chicken meat collected from three supershops of Dhaka city (5). However, both studies sampled frozen chicken meat from a limited number of supershops (three to five) of Dhaka city only, and the multidrug resistance pattern of ESBL-producing *Salmonella* was not investigated.

Therefore, an inclusive study is required to have a complete picture of ESBL-producing *Salmonella* contamination and resistance patterns in frozen chicken meat of different supershops across different divisional megacities in Bangladesh. The present study was, therefore, conducted to determine the prevalence of ESBL-producing *Salmonella* and their AMR pattern isolated from frozen chicken meat in Bangladesh. We also focused on the determination of β -lactamase- and PMQR-encoding genes in *Salmonella* from frozen chicken meat, which has not yet been investigated in Bangladesh.

MATERIALS AND METHODS

Sample collection. A total of 113 domestic frozen chicken meat samples (82 broilers, 31 cockerels) were purchased from 40 outlets of nine different supershops in five divisional megacities of Bangladesh (Fig. 1) during April to December 2019. Each supershop had its own branded frozen chicken meat. It was noted that all the supershops collected their live chickens from contract farmers, that is, the supershop authority had a contract with the poultry farmers: they provided the chickens, feed, veterinary care, and technical advice, while the farmers provided the day-to-day care of the birds, land, and housing, as well as utilities and maintenance of the housing. Meat samples included whole chicken or meat cuts (breast, drumstick, leg, and wing muscle). After purchase, samples were directly placed into a sterilized plastic bag in a cooler box with ice packs and were transported to the laboratory and processed on the same day. Data on brand name, source of chicken, processing and packaging of meat, and special labels (e.g., green or organic chickens) were also collected.

***Salmonella* isolation and confirmation.** *Salmonella* isolation and identification were carried out according to the ISO 6579-1:2017(en) standard, with some modifications (16, 28). For the preenrichment of *Salmonella*, 25 g of meat sample was homogenized with 225 mL of buffered peptone water and incubated at 37°C for 18 to 24 h. An amount of 0.1 mL of the

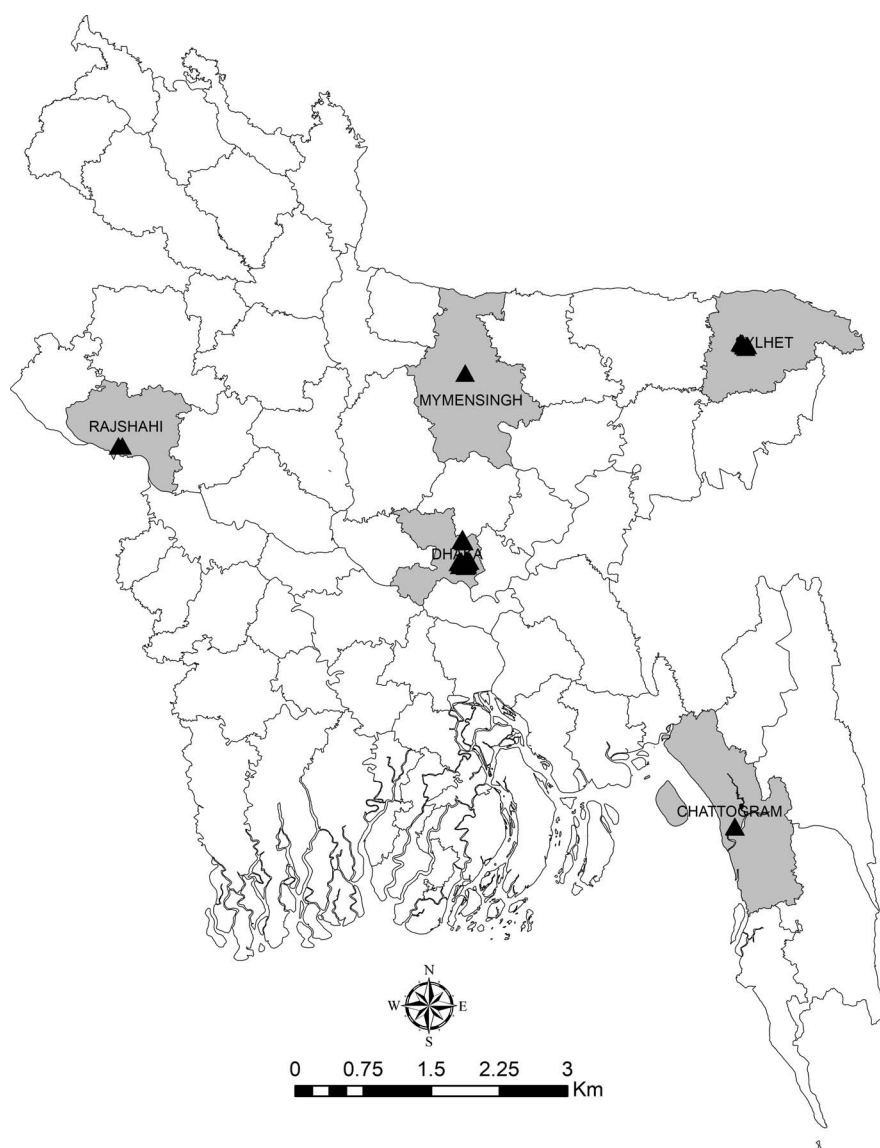


FIGURE 1. Map showing sampling sites (▲) in five divisional megacities (high-lighted areas) of Bangladesh.

culture was then mixed with 10 mL of Rappaport-Vassiliadis Soya broth and incubated for 24 h at 37°C. A loopful of culture broth was then streaked onto xylose lysine desoxycholate (XLD) agar and incubated at 37°C for 24 h. Typical colonies of *Salmonella*, with a black center and a slightly transparent zone of reddish color, were subjected to Gram stain and biochemical assays, which included catalase, oxidase, indole, methyl red test, Voges-Proskauer test, and sugar fermentation test using triple sugar iron agar. All presumptive *Salmonella* isolates were then subjected to DNA extraction using a hot-cold lysis method (49). Briefly, a loopful of culture from XLD agar was mixed into 200 µL of double-distilled water in a sterile 1.5-mL Eppendorf tube to obtain a turbid suspension of bacteria. The bacterial suspension was kept in a boiling water bath for 15 min, immediately cooled at –20°C for 15 min, and centrifuged at $18,000 \times g$ for 5 min. The supernatant containing genomic DNA was collected in a new tube and was used as a DNA template for PCR reactions.

A uniplex PCR targeting the ITS (312 bp) gene was used for the confirmation of the *Salmonella* genus (13). Primers used were ITSF (5'-TATAGCCCCATCGTGTCAGAAC-3') and ITSR (5'-TGCGGCTGGATCACCTCCTT-3'). The PCR reaction mixture (25 µL) consisted of 12.5 µL of OneTaq Quick-Load 2× PCR master mix with standard buffer (New England BioLabs Inc.,

Ipswich, MA), 1.5 µL (15 pmol) each of forward and reverse primers, 1 µL of DNA template, and 8.5 µL of nuclease-free water. The thermal profile consisted of initial denaturation at 94°C for 5 min, 35 cycles of separation at 94°C for 30 s, primer annealing at 60°C for 30 s, and extension at 72°C for 50 s with a final elongation at 72°C for 10 min. The PCR products were gel electrophoresed (1.5% agarose gel containing ethidium bromide) and were visualized under a UV transilluminator and photographed. All PCR-confirmed *Salmonella* isolates were stored on nutrient broth containing 50% (v/v) glycerol at –20°C for further study.

Antimicrobial susceptibility testing. The AMR profile of all isolates was determined using the Kirby-Bauer disk diffusion method as described by the Clinical and Laboratory Standards Institute (CLSI) (14). A panel of 36 antimicrobials representing 15 different antimicrobial classes were used for antimicrobial susceptibility testing as stated below.

(i) Fluoroquinolones: nalidixic acid (30 µg), ciprofloxacin (5 µg), levofloxacin (5 µg), norfloxacin (10 µg), ofloxacin (5 µg), gatifloxacin (5 µg), pefloxacin (5 µg)

TABLE 1. Oligonucleotide primers used for the detection of β -lactamase- and PMQR-encoding genes^a

Genes	Primers	Sequence 5'→3'	Size (bp)	Reference
BSBL-encoding genes				36
<i>bla</i> _{TEM}	TEM-410F TEM-781R	GGTCGCCGCATACACTATTCTC TTTATCCGCCTCCATCCAGTC	372	
<i>bla</i> _{SHV}	SHV-287F SHV-517R	CCAGCAGGATCTGGTGGACTAC CCGGGAAGCGCCTCAT	231	
ESBL-encoding genes				36
<i>bla</i> _{CTX-M-1}	ctxm1-15F ctxm1-02R	GAATTAGAGCGGCAGTCGGG CACAACCCAGGAAGCAGGC	588	
<i>bla</i> _{CTX-M-2}	ctxm2-39F ctxm2-45R	GATGGCGACGTACCCC CAAGCCGACCTCCCGAAC	107	
Carbapenemase-encoding gene				29
<i>bla</i> _{NDM-1}	NDM-1-F NDM-1-R	CTTCCAACGGTTTGATCGTC TAGTGCTCAGTGTCCGCATC	465	
PMQR-encoding genes				52
<i>qnrA</i>	qnrA-F qnrA-R	ATTCTCACGCCAGGATTG GATCGGCAAAGGTTAGGTCA	516	
<i>qnrB</i>	qnrB-F qnrB-R	GATCGTGAAAGCCAGAAAGG ACGATGCCTGGTAGTTGTCC	469	
<i>qnrS</i>	qnrS-F qnrS-R	ACGACATTCGTCAACTGCAA TAAATTGGCACCCGTGAGGC	417	

^a BSBL, broad-spectrum β -lactamase; ESBL, extended-spectrum β -lactamase; *bla*_{NDM}, New Delhi metallo- β -lactamase; PMQR, plasmid-mediated quinolones resistance.

(ii) Nonextended spectrum cephalosporins. First-generation cephalosporins: cephalexin (30 μ g), cephadrine (30 μ g). Second-generation cephalosporins: cefuroxime (30 μ g), cefaclor (30 μ g)

(iii) Extended-spectrum cephalosporins. Third-generation cephalosporins: cefotaxime (30 μ g), ceftriaxone (30 μ g), ceftazidime (30 μ g), cefixime (5 μ g). Fourth-generation cephalosporins: cefepime (30 μ g)

(iv) Cephamycins: cefoxitin (30 μ g)

(v) Penicillins: ampicillin (10 μ g), amoxicillin (10 μ g)

(vi) Penicillins plus β -lactamase inhibitors: amoxicillin-clavulanic acid (30 μ g)

(vii) Antipseudomonal penicillins plus β -lactamase inhibitors: piperacillin-tazobactam (110 μ g)

(viii) Carbapenems: imipenem (10 μ g), meropenem (10 μ g)

(xi) Monobactams: aztreonam (30 μ g)

(x) Aminoglycosides: gentamicin (10 μ g), amikacin (30 μ g), streptomycin (10 μ g), neomycin (30 μ g), tobramycin (10 μ g)

(xi) Tetracyclines: tetracycline (30 μ g), oxytetracycline (30 μ g), doxycycline (10 μ g)

(xii) Folate pathway inhibitors: trimethoprim-sulfamethoxazole (25 μ g)

(xiii) Glycylcyclines: tigecycline (15 μ g)

(xiv) Phenolics: chloramphenicol (30 μ g)

(xv) Macrolides: azithromycin (15 μ g)

Salmonella strains grown for 24 h in XLD agar were diluted in sterile nutrient broth and adjusted to an opacity equivalent to a 0.5 McFarland turbidity standard. Bacterial suspensions were spread on Mueller-Hinton agar. The antimicrobial discs were placed over the lawn and incubated at 37°C for 18 to 24 h. The diameter of zone of inhibition was measured to the nearest whole millimeter using a ruler. The interpretive category for each isolate (susceptible, intermediate, and resistant) was determined according to the CLSI guidelines (14). Isolates resistant to one or more agent in three or more antimicrobial classes were classed as MDR,

and isolates resistant to one or more agent in all but two or fewer antimicrobial classes were categorized as extensively drug resistant (40). Antimicrobial discs were procured from Biomaxima (Lublin, Poland); however, oxytetracycline and streptomycin were purchased from Oxoid (Basingstoke, UK).

ESBL-producing *Salmonella* screening. ESBL production by the *Salmonella* isolates was screened by using the double-disk synergy method (17). A disc of amoxicillin-clavulanic acid (30 μ g) was placed on the center of the Mueller-Hinton agar plate, which was previously inoculated with *Salmonella* isolate. Then, each disc of ceftriaxone (30 μ g), cefotaxime (30 μ g), and ceftazidime (30 μ g) was placed around the amoxicillin-clavulanic acid disc at 30 mm apart and was incubated at 37°C for 18 to 24 h. The phenotypic presence of ESBL was determined by the observation of a clear extension of the edge of the inhibition zone of any of the antibiotics toward the central disc (amoxicillin-clavulanic acid). The isolate was considered positive for ESBL if the zone diameters of inhibition for ceftazidime, cefotaxime, and ceftriaxone were, respectively, ≤ 22 , ≤ 27 , and ≤ 25 mm (14).

Detection of resistance genes. Two separate multiplex PCR assays targeting β -lactamase-encoding genes (BSBL: *bla*_{TEM}, *bla*_{SHV}; ESBL: *bla*_{CTX-M-1}, *bla*_{CTX-M-2}), and PMQR genes (*qnrA*, *qnrB*, and *qnrS*) were standardized and used in our study (36, 52). In addition, a uniplex PCR assay was used to detect carbapenemase-encoding gene (*bla*_{NDM-1}) (29). Amplification was carried out using the primers listed in Table 1. For β -lactamase-encoding genes, an aliquot of 1 μ L of template DNA was added to 25 μ L of PCR mixture containing 1.0 μ L (10 pmol) of each of the forward and reverse primers, 12.5 μ L of PCR master mix (Thermo Fisher Scientific, Waltham, MA), and 3.5 μ L of nuclease-free water. Amplification was performed by using a Veriti 96-well thermal cycler (Thermo Fisher Scientific) with this thermal profile: heating at 95°C for 5 min, followed by

TABLE 2. Prevalence of *Salmonella* isolated from frozen chicken meat of different brands^a

Variable	Total no. of samples	No. (%) of <i>Salmonella</i> -positive isolates	ESBL- <i>Salmonella</i> , no. (%)	Non-ESBL- <i>Salmonella</i> , no. (%)
Brand^b				
Brand 1	23	18 (78.3)	10 (55.6) A	8 (44.4) A
Brand 2	40	28 (70.0)	14 (50.0) A	14 (50.0) A
Brand 3	28	18 (64.3)	12 (66.7) A	6 (33.3) A
Brand 4	8	4 (50.0)	1 (25.0) A	3 (75.0) A
Brand 5	2	1 (50.0)	1 (100.0) A	0
Brand 6	2	1 (50.0)	1 (100.0) A	0
Brand 7	5	2 (40.0)	2 (100.0) A	0
Brand 8	3	0	0	0
Brand 9	2	2 (100.0)	2 (100.0) A	0
Division				
Dhaka	82	57 (69.5)	31 (54.4) A	26 (45.6) A
Chattogram	10	5 (50.0)	4 (80.0) A	1 (20.0) A
Sylhet	11	9 (81.8)	5 (55.6) A	4 (44.4) A
Mymensingh	5	2 (40.0)	2 (100.0) A	0
Rajshahi	5	1 (20.0)	1 (100.0) A	0
Chicken type				
Broiler	82	56 (68.3)	32 (57.1) A	24 (42.9) A
Cockerel	31	18 (58.1)	11 (61.1) A	7 (38.9) A
Production type				
Organic	10	5 (50.0)	1 (20.0) A	4 (80.0) A
Nonorganic	103	69 (67.0)	42 (60.9) A	27 (39.1) A
Meat type				
Breast	27	17 (63.0)	9 (52.9) A	8 (47.1) A
Drumstick	30	21 (70.0)	15 (71.4) A	6 (28.6) A
Leg	3	2 (66.7)	2 (100.0) A	0
Wing	19	15 (78.9)	6 (40.0) A	9 (60.0) A
Whole chicken	34	19 (55.9)	11 (57.9) A	8 (42.1) A
Total	113	74 (65.5)	43 (58.1)	31 (41.9)

^a ESBL-*Salmonella*, ESBL-producing *Salmonella*; non-ESBL-*Salmonella*, non-ESBL-producing *Salmonella*. Values in the same column followed by the same letter do not differ significantly ($P > 0.05$).

^b Name of the frozen chicken meat, which was named after the name of the respective supershop.

25 cycles of denaturation at 95°C for 30 s, primer annealing at 60°C for 1 min, extension at 72°C for 1 min, and a final extension step for 10 min at 72°C. For the multiplex assay targeting the PMQR genes, PCR was performed with a 25-μL PCR mixture that contained 1.5 μL (15 pmol) of each of the forward and reverse primers, 12.5 μL of PCR master mix (Thermo Fisher Scientific), 1 μL of DNA template, 2.5 μL of nuclease-free water, and with the following cycling parameters: an initial denaturation step at 94°C for 5 min; 30 cycles of 94°C for 45 s, 57°C for 45 s, and 72°C for 1 min; and a final extension step at 72°C for 8 min. For carbapenemase-encoding gene, PCR was carried out with 25 μL of reaction volume, which contained 1.5 μL (15 pmol) of each primer, 12.5 μL of PCR master mix (Thermo Fisher Scientific), 1 μL of template DNA, and 8.5 μL of nuclease-free water. PCR was run with an initial denaturation at 95°C for 7 min; 30 cycles of 95°C for 1 min, 55°C for 1 min, and 68°C for 1 min; and a final elongation at 68°C for 7 min. Positive (containing strains with known β-lactamase and *qnr* genes) and negative (sterile phosphate buffer saline) controls were included in each run. PCR amplicons were visualized by using a UV transilluminator and were photographed after electrophoresis on 1.5% agarose gel containing 0.5 μg/mL of ethidium bromide.

Statistical analysis. Descriptive statistics were used to compute the prevalence of *Salmonella* and resistance percentage. The prevalence of *Salmonella* among different brands, sampling area, chicken types, production types, and meat types, etc., were determined using the Z test for proportions and Fisher's exact test (wherever appropriate). The chi-square test was used to compare the resistance rate and carriage of β-lactamase- and PMQR-encoding genes in *Salmonella* isolated from frozen chicken meat. An alpha of ≤0.05 was used for statistical significance. The statistical software package SPSS version 22.0 (IBM, Somers, NY) was used for the analyses.

RESULTS

Prevalence and distribution of *Salmonella* isolates.

The prevalence of *Salmonella* isolated from frozen chicken meat is summarized in Table 2. Seventy-four (65.5%) *Salmonella* isolates were recovered from 113 frozen chicken meat samples. All *Salmonella* isolates tested positive for the *Salmonella*-specific ITS gene. Of the 74 *Salmonella* isolates, 43 (58.1%) were identified as ESBL-producing *Salmonella* (ESBL-*Salmonella*), and the remaining 31 (41.9%) isolates were non-ESBL-producing *Salmonella*.

TABLE 3. Distribution of *Salmonella* isolates resistant to multiple classes of antimicrobials^a

Variable	No. (%) of isolates resistant to multiple antimicrobial classes			
	3–5	6–8	9–11	12–15
Brand				
Brand 1	6 (33.3) AB	10 (55.6) A	2 (11.1) A	0
Brand 2	16 (57.1) B	8 (28.6) A	3 (10.7) A	1 (3.6) A
Brand 3	4 (22.2) A	8 (44.4) A	6 (33.3) A	0
Brand 4	2 (50.0) AB	1 (25.0) A	1 (25.0) A	0
Brand 5	0	1 (100.0) A	0	0
Brand 6	1 (100.0) AB	0	0	0
Brand 7	0	1 (50.0) A	1 (50.0) A	0
Brand 9	0	1 (50.0) A	0	0
Division				
Dhaka	23 (40.4) A	24 (42.1) A	9 (15.8) B	1 (1.8) A
Chattogram	0	2 (40.0) A	3 (60.0) A	0
Sylhet	6 (66.7) B	3 (33.3) A	0	0
Mymensingh	0	1 (50.0) A	1 (50.0) A	0
Rajshahi	1 (100.0) B	0	0	0
Chicken type				
Broiler	25 (44.6) A	19 (33.9) A	11 (19.6) A	1 (1.8) A
Cockerel	5 (27.8) A	11 (61.1) B	2 (11.1) A	0
Production type				
Organic	2 (40.0) A	2 (40.0) A	1 (20.0) A	0
Nonorganic	28 (40.6) A	28 (40.6) A	12 (17.4) A	1 (1.4) A
Meat type				
Breast	5 (29.4) A	8 (47.1) A	3 (17.6) A	1 (5.9)
Drumstick	8 (38.1) A	10 (47.6) A	3 (14.3) A	0
Leg	0	1 (50.0) A	1 (50.0) A	0
Wing	11 (73.3) B	0	4 (26.7) A	0
Whole chicken	6 (31.6) A	11 (57.9) A	2 (10.5) A	0
Total	30 (40.5)	30 (40.5)	13 (17.6)	1 (1.4)

^a Values in the same column followed by different letters differ significantly ($P \leq 0.05$). Samples from brand 8 were negative for *Salmonella*.

(non-ESBL-*Salmonella*). Among the eight brands, the prevalence of ESBL-*Salmonella* among different brands varied from 25 to 100%, whereas the prevalence of non-ESBL-*Salmonella* varied from 33.3 to 75%. No significant difference was observed in the prevalence of ESBL-*Salmonella* and non-ESBL-*Salmonella* among brands. None of the *Salmonella* isolates were recovered from frozen chicken meat of brand no. 8.

Table 2 also shows the distribution of *Salmonella* related to divisions, chicken types, production types, and meat types. The prevalence of *Salmonella* was highest in Sylhet division (81.8%), followed by Dhaka division (69.5%); the lowest prevalence was found in Rajshahi division (20%). Neither chicken type (broiler or cockerel) nor production type (organic or nonorganic) had a significant effect on the prevalence of *Salmonella*. *Salmonella* prevalences in broiler and cockerel chickens were 68.3 and 58.1%, respectively; in organic and nonorganic production types, the prevalences were 50 and 67%, respectively. Distribution by meat type revealed that approximately 80% of wing muscle samples and 70% of drumstick muscle samples were contaminated with *Salmo-*

nella. However, the prevalence of ESBL-*Salmonella* and non-ESBL-*Salmonella* did not differ significantly among different variables.

AMR profile. Of the 74 *Salmonella* isolates, all were MDR. Overall, 40.5% isolates were resistant to three to five antimicrobial classes, 40.5% to six to eight classes, 17.6% to 9 to 11 classes, and 1.4% to 12 to 15 classes (Table 3). Notably, one isolate (1.4%) was possibly extensively drug resistant; it showed resistance to 12 of 15 antimicrobial classes. When the MDR pattern was analyzed by brand, the majority of the isolates from all brands showed resistance to three to five and six to eight antimicrobial classes. Only one isolate from brand 2 was resistant to 12 to 15 classes of antimicrobials. Analysis by division revealed that the highest resistance to three to five antimicrobial classes was observed in Rajshahi (100%), followed by Sylhet (66.7%) divisions, whereas resistance to 9 to 11 classes was higher in Chattogram division (60%). Resistance to 12 to 15 classes was observed in one isolate (1.8%) from Dhaka division only. As for chicken and production types, the MDR pattern was similar among *Salmonella* isolates from

TABLE 4. Brand-wise distribution of *Salmonella* isolates resistant to multiple antimicrobial agents^a

Brand	No. (%) resistant isolates to indicated no. of antimicrobials			
	3–7	8–12	13–17	18–22
Brand 1	4 (22.2) A	9 (50.0) AB	1 (5.6) A	4 (22.2) A
Brand 2	12 (42.9) A	11 (39.3) AB	3 (10.7) A	2 (7.1) A
Brand 3	3 (16.7) A	5 (27.8) B	7 (38.9) B	3 (16.7) A
Brand 4	2 (50.0) A	1 (25.0) AB	0	1 (25.0) A
Brand 5	0	0	1 (100.0) C	0
Brand 6	0	1 (100.0) A	0	0
Brand 7	0	1 (50.0) AB	1 (50.0) B	0
Brand 9	0	2 (100.0) A	0	0
Total	21 (28.4)	30 (40.5)	13 (17.6)	10 (13.5)

^a Values in the same column followed by different letters differ significantly ($P \leq 0.05$). Samples from brand 8 were negative for *Salmonella*.

broiler and cockerel chickens and from organic and nonorganic production types. According to distribution by meat type, high percentages (73.3%) of isolates recovered from wing muscles were resistant to three to five antimicrobial classes. Resistance to six to eight antimicrobial classes was significantly higher in whole chicken (57.9%) than in other types of meat.

Among the 74 *Salmonella* isolates, all were resistant to at least three, and up to 22, of 36 antimicrobial agents tested (Table 4). Overall resistance to 8 to 12 antimicrobials was observed in 40.5% of isolates. In addition, 28.4% of isolates were resistant to three to seven antimicrobials, and 17.6% to 13 to 17 antimicrobials (Table 4). Remarkably, 13.5% of isolates showed resistance to 20 to 22 antimicrobials. Significant differences were found in the resistance to antimicrobials among brands.

Overall, the highest individual AMR was observed against oxytetracycline (100%). Resistance to trimethoprim-sulfamethoxazole (89.2%), tetracycline (86.5%), nalidixic acid (83.8%), amoxicillin (74.3%), and pefloxacin (70.3%) were also high (Table 5). Low resistance (1.4 to 4.1%) was found to ceftazidime, cefepime, cefuroxime, ceftriaxone, cefotaxime, aztreonam, and tigecycline. Note that the resistance to carbapenems (imipenem) was 48.6%. The variation in the resistance pattern between ESBL-*Salmonella* and non-ESBL-*Salmonella* isolates was determined: among ESBL-*Salmonella* isolates, high resistance was observed to oxytetracycline (100%), tetracycline (90.7%), and trimethoprim-sulfamethoxazole and amoxicillin (81.4%); in non-ESBL-*Salmonella* isolates, high resistance was observed to oxytetracycline and trimethoprim-sulfamethoxazole (100.0%), nalidixic acid (90.3%), and tetracycline (80.6%) (Table 5). However, no significant differences were observed in resistance to these antimicrobial agents between ESBL and non-ESBL-*Salmonella*, with the exception of amoxicillin.

Prevalence of resistance genes. As for BSBL-encoding genes, all the isolates were positive for the *bla*_{TEM}

gene (Table 6 and Fig. 2). None of the isolates had the *bla*_{SHV} gene. As for ESBL-encoding genes, two (4.7%) *Salmonella* isolates harbored the *bla*_{CTX-M-1} gene (Table 6 and Fig. 2). The *bla*_{CTX-M-2} gene was not detected in any of the isolates. As for the carbapenemase-encoding gene, 15 (20.3%) isolates carried *bla*_{NDM-1} (Table 6 and Fig. 3). Among PMQR genes, *qnrS* and *qnrA* were detected in five (6.8%) and three (4.1%) *Salmonella* isolates, respectively (Table 6 and Fig. 4). None of the isolates carried *qnrB*.

The association between AMR phenotypes and resistance genes demonstrated that not only did the *Salmonella* isolates resistant to cephalosporin and fluoroquinolones harbor the respective β -lactamase and PMQR genes but that the *Salmonella* isolates nonresistant to those antimicrobials also carried these resistance genes (Table 7). *Salmonella* isolates resistant to antimicrobials other than cephalosporins, such as cephamycins, penicillins, penicillins- β -lactamase inhibitors, antipseudomonal penicillins- β -lactamase inhibitors, and carbapenems, also harbored the ESBL-encoding gene (*bla*_{CTX-M-1}). We observed that none of the isolates resistant to cefuroxime, ceftazidime, cefixime, and cefepime harbored the *bla*_{CTX-M-1} gene, whereas certain percentages of the isolates nonresistant to these antimicrobials did carry the *bla*_{CTX-M-1} gene. The occurrence of carbapenemase-encoding gene (*bla*_{NDM-1}) in *Salmonella* isolates resistant to imipenem and meropenem was 36.1 and 30%, respectively. A number of isolates that were resistant to cephalosporins, cephamycins, penicillins, penicillins- β -lactamase inhibitors, and antipseudomonal penicillins- β -lactamase inhibitors also contained the *bla*_{NDM-1} gene. Furthermore, the very few *Salmonella* isolates that were resistant to the antimicrobials of the fluoroquinolones class contained either *qnrA* or *qnrS* or both genes. Similarly, a few of the isolates that were nonresistant to fluoroquinolone antimicrobials also harbored either *qnrA* or *qnrS* or both genes.

DISCUSSION

To our knowledge, the present study reports the first and quite comprehensive findings on the extent and distribution of ESBL-*Salmonella* and their AMR pattern including resistance genes (BSBL-, ESBL-, carbapenemase-, and PMQR-encoding) in frozen chicken meat collected from almost all branded supershops located in five divisional megacities of Bangladesh. The findings of the present study demonstrate a high prevalence of *Salmonella* in frozen chicken meat (65.5%); this is inconsistent with a previous report of Uddin et al. (66), who observed that the bacterial load of *Salmonella* was lower than those of *E. coli* and *Shigella* spp. in frozen chicken meat. However, a variable prevalence of *Salmonella* (16 to 57%) in raw chicken meat from live bird markets was reported earlier in Bangladesh (6, 7, 18, 21, 34, 51). Many factors, such as the origin of chicken meat, sample types, sample number, isolation and testing methods, handler's hygiene, slaughterhouse sanitation, processing and packaging of chicken meat, and other practices through to the food chain, may make comparisons of different studies difficult (70). The current study reports higher prevalences of *Salmonella* in Sylhet and Dhaka

TABLE 5. Individual antimicrobial resistance pattern of *Salmonella* isolated from frozen chicken meat

Antimicrobial category	No. (%) of isolates ^a								
	Overall (n = 74)			ESBL- <i>Salmonella</i> (n = 43)			Non-ESBL- <i>Salmonella</i> (n = 31)		
	R	I	S	R	I	S	R	I	S
Fluoroquinolones									
Ciprofloxacin	28 (37.8)	38 (51.4)	8 (10.8)	19 (44.2) A	19 (44.2)	5 (11.6)	9 (29.0) A	19 (61.3)	3 (9.7)
Nalidixic acid	62 (83.8)	5 (6.8)	7 (9.5)	34 (79.1) A	4 (9.3)	5 (11.6)	28 (90.3) A	1 (3.2)	2 (6.5)
Levofloxacin	8 (10.8)	9 (12.2)	57 (77.0)	4 (9.3) A	5 (11.6)	34 (79.1)	4 (12.9) A	4 (12.9)	23 (74.2)
Norfloxacin	14 (18.9)	17 (23.0)	43 (58.1)	11 (25.6) A	12 (27.9)	20 (46.5)	3 (9.7) B	5 (16.1)	23 (74.2)
Gatifloxacin	22 (29.7)	10 (13.5)	42 (56.8)	10 (23.3) A	7 (16.3)	26 (60.5)	12 (38.7) B	3 (9.7)	16 (51.6)
Pefloxacin	52 (70.3)	0	22 (29.7)	29 (67.4) A	0	14 (32.6)	23 (74.2) A	0	8 (25.8)
Ofloxacin	26 (35.1)	11 (14.9)	37 (50.0)	11 (25.6) A	9 (20.9)	23 (53.5)	15 (48.4) B	2 (6.5)	14 (45.2)
Nonextended spectrum cephalosporins									
1st generation									
Cephalexin	7 (9.5)	0	67 (90.5)	7 (16.3)	0	36 (83.7)	0	0	31 (100.0)
Cephadrine	9 (12.2)	0	65 (87.8)	9 (20.9)	0	34 (79.1)	0	0	31 (100.0)
2nd generation									
Cefuroxime	2 (2.7)	1 (1.4)	71 (95.9)	2 (4.7)	1 (2.3)	40 (93.0)	0	0	31 (100.0)
Cefaclor	10 (13.5)	1 (1.4)	63 (85.1)	8 (18.6) A	1 (2.3)	34 (79.1)	2 (6.5) B	0	29 (93.5)
Extended-spectrum cephalosporins									
3rd generation									
Cefixime	4 (5.4)	1 (1.4)	69 (93.2)	2 (4.7) A	1 (2.3)	40 (93.0)	2 (6.5) A	0	29 (93.5)
Ceftazidime	1 (1.4)	4 (5.4)	69 (93.2)	1 (2.3)	4 (9.3)	38 (88.4)	0	0	31 (100.0)
Ceftriaxone	2 (2.7)	2 (2.7)	70 (94.6)	2 (4.7)	2 (4.7)	39 (90.7)	0	0	31 (100.0)
Cefotaxime	3 (4.1)	19 (25.7)	52 (70.3)	3 (7.0)	19 (44.2)	21 (48.8)	0	0	31 (100.0)
4th generation									
Cefepime	1 (1.4)	22 (29.7)	51 (68.9)	1 (2.3)	18 (41.9)	24 (55.8)	0	4 (12.9)	27 (87.1)
Cephameycins									
Cefoxitin	6 (8.1)	1 (1.4)	67 (90.5)	6 (14.0)	1 (2.3)	36 (83.7)	0	0	31 (100.0)
Penicillins									
Ampicillin	47 (63.5)	1 (1.4)	26 (35.1)	31 (72.1) A	0	12 (27.9)	16 (51.6) A	1 (3.2)	14 (45.2)
Amoxicillin	55 (74.3)	0	19 (25.7)	35 (81.4) A	0	8 (18.6)	11 (35.5) B	0	20 (64.5)
Penicillins + β-lactamase inhibitors									
Amoxicillin-clavulanic acid									
	19 (25.7)	16 (21.6)	39 (52.7)	17 (39.5) A	12 (27.9)	14 (32.6)	2 (6.5) B	4 (12.9)	25 (80.6)
Antipseudomonal penicillins + β-lactamase inhibitors									
Piperacillin-tazobactam									
	15 (20.3)	14 (18.9)	45 (60.8)	10 (23.3) A	14 (32.6)	19 (44.2)	5 (16.1) A	0	26 (83.9)
Carbapenems									
Imipenem	36 (48.6)	11 (14.9)	27 (36.5)	25 (58.1) A	7 (16.3)	11 (25.6)	11 (35.5) B	4 (12.9)	16 (51.6)
Meropenem	10 (13.5)	14 (18.9)	50 (67.6)	7 (16.3) A	5 (11.6)	31 (72.1)	3 (9.7) A	9 (29.0)	19 (61.3)
Monobactams									
Aztreonam	3 (4.1)	4 (5.4)	67 (90.5)	3 (7.0)	4 (9.3)	36 (83.7)	0	0	31 (100)
Aminoglycosides									
Gentamicin	16 (21.6)	5 (6.8)	53 (71.6)	12 (27.9) A	1 (2.3)	30 (69.8)	4 (12.9) B	4 (12.9)	23 (74.2)
Tobramycin	10 (13.5)	5 (6.8)	59 (79.7)	8 (18.6) A	4 (9.3)	31 (72.1)	2 (6.5) B	1 (3.2)	28 (90.3)
Amikacin	14 (18.9)	36 (48.6)	24 (32.4)	11 (25.6) A	16 (37.2)	16 (37.2)	3 (9.7) B	20 (64.5)	8 (25.8)
Streptomycin	23 (31.1)	22 (29.7)	29 (39.2)	15 (34.9) A	15 (34.9)	13 (30.2)	8 (25.8) A	7 (22.6)	16 (51.6)
Neomycin	26 (35.1)	22 (29.7)	26 (35.1)	16 (37.2) A	12 (27.9)	15 (34.9)	10 (32.3) A	10 (32.3)	11 (35.5)
Tetracyclines									
Tetracycline	64 (86.5)	8 (10.8)	2 (2.7)	39 (90.7) A	4 (9.3)	0	25 (80.6) A	4 (12.9)	2 (6.5)
Oxytetracycline	74 (100.0)	0	0	43 (100.0) A	0	0	31 (100.0) A	0	0
Doxycycline	43 (58.1)	20 (27.0)	11 (14.9)	20 (46.5) A	17 (39.5)	6 (14.0)	23 (74.2) B	3 (9.7)	5 (16.1)

TABLE 5. Continued

Antimicrobial category	No. (%) of isolates ^a								
	Overall (n = 74)			ESBL- <i>Salmonella</i> (n = 43)			Non-ESBL- <i>Salmonella</i> (n = 31)		
	R	I	S	R	I	S	R	I	S
Folate pathway inhibitors									
Trimethoprim-sulfamethoxazole	66 (89.2)	0	8 (10.8)	35 (53.0) A	0	8 (18.6)	31 (100.0) A	0	0
Glycylcyclines									
Tigecycline	3 (4.1)	11 (14.9)	60 (81.1)	3 (7.0)	7 (16.3)	33 (76.7)	0	4 (12.9)	27 (87.1)
Phenicol									
Chloramphenicol	21 (28.4)	14 (18.9)	39 (52.7)	16 (37.2) A	7 (16.3)	20 (46.5)	5 (16.1) B	7 (22.6)	19 (61.3)
Macrolids									
Azithromycin	34 (45.9)	0	40 (54.1)	21 (48.8) A	0	22 (51.2)	13 (41.9) A	0	18 (58.1)

^a n, number of isolates; R, resistant; I, intermediate; S, susceptible; ESBL-*Salmonella*, ESBL-producing *Salmonella*; non-ESBL-*Salmonella*, non-ESBL-producing *Salmonella*. Values in the same row followed by different letters differ significantly ($P \leq 0.05$).

divisions than in Rajshahi; this is in accordance with an earlier report in which a relatively higher percentage of *Salmonella* contamination was observed in chicken meat in Dhaka division of Bangladesh (31). However, the production category (organic versus nonorganic) and chicken types (broiler versus cockerel) had minimal effects on *Salmonella* contamination. In the present study, different types of meat samples had variable degrees of contamination with *Salmonella*. The intact muscle tissues of healthy slaughtered chickens are mostly sterile, but they may be contaminated with *Salmonella* during the evisceration step, which may further facilitate cross-contamination during subsequent processing, cut-up, and preparation activities (62).

TABLE 6. Prevalence of resistance genes in *Salmonella* isolated from frozen chicken meat^a

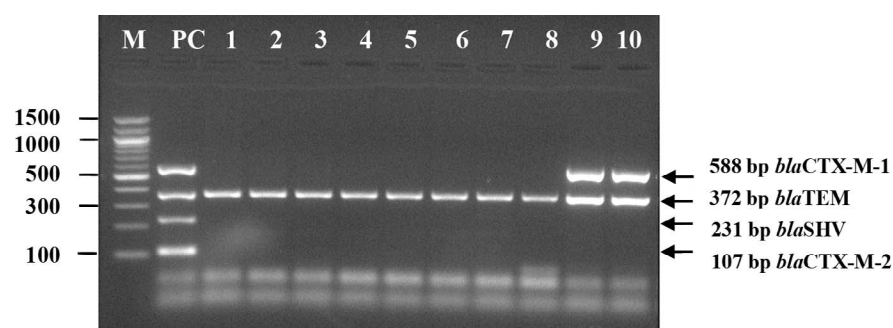
Resistance genes	Total (n = 74)	ESBL- <i>Salmonella</i> (n = 43)	Non-ESBL- <i>Salmonella</i> (n = 31)
BSBL-encoding genes			
<i>bla</i> _{TEM}	74 (100.0)	43 (100.0)	31 (100.0)
<i>bla</i> _{SHV}	0	0	0
ESBL-encoding genes			
<i>bla</i> _{CTX-M-1}	2 (2.7)	2 (4.7)	0
<i>bla</i> _{CTX-M-2}	0	0	0
Carbapenemase-encoding gene			
<i>bla</i> _{NDM-1}	15 (20.3)	10 (23.3)	5 (16.1)
PMQR-encoding genes			
<i>qnrA</i>	3 (4.1)	2 (4.7)	1 (3.2)
<i>qnrB</i>	0	0	0
<i>qnrS</i>	5 (6.8)	4 (9.3)	1 (3.2)

^a n, number of isolates; ESBL-*Salmonella*, ESBL-producing *Salmonella*; non-ESBL-*Salmonella*, non-ESBL-producing *Salmonella*; BSBL, broad-spectrum β -lactamase; ESBL, extended-spectrum β -lactamase; *bla*_{NDM}, New Delhi metallo- β -lactamase; PMQR, plasmid-mediated quinolone resistance.

The contamination of chicken meat with ESBL-*Salmonella* has recently become a worldwide concern. In the present study, the isolation rate of ESBL-*Salmonella* from frozen chicken meat was 58.1%, which can be compared with the findings of a similar study conducted in Nepal, in which the authors reported that 55.2% of frozen chicken meat samples were contaminated with ESBL-*Salmonella* (61). This study demonstrated that the prevalence of ESBL-*Salmonella* varied from brand to brand, perhaps due to differences in the hygiene and sanitation levels of each brand. Several factors are responsible for ensuring the microbiological quality of frozen chicken meat, including maintenance of appropriate cold chain storage from the processing to the retail level and to the consumer and also sanitation during handling at the retail end (57).

AMR in *Salmonella* has become a notable public health concern globally. Surveillance data have documented a distinguishable rise in overall AMR among *Salmonella* species, from 20 to 30% in the early 1990s to a peak of 70% in some countries in the 2000s (64). The results demonstrate that 100% of the isolates were resistant to at least three antimicrobial agents, and 13.5% were resistant to more than 20 antimicrobials. A high percentage of antimicrobial-resistant *Salmonella* from frozen chicken meat was also reported by several investigators (2, 68, 71). The high prevalence of antimicrobial-resistant *Salmonella* in frozen chicken meat was associated with the brands, which could be due to brand-level differences in production and processing, including variations in the use of antimicrobials. We observed that all the *Salmonella* isolates were MDR and that the majority of the isolates showed resistance to 9 to 11 and 12 to 15 classes of antimicrobials; this is in line with some previous observations on *Salmonella* contamination in retail chicken meat (6, 60, 71) but differs from some other reports (31, 68). Of note, the present study also observed that 1.4% of isolates were possibly extensively drug resistant. Recently, a report from Pakistan described the emergence of an extensively drug-resistant *Salmonella* Typhi clone in humans (35). The high frequency of MDR

FIGURE 2. *BSBL- and ESBL-encoding genes of Salmonella isolates from frozen chicken meat detected by multiplex PCR.* M, 100-bp DNA ladder; PC, positive control; lanes 1 to 10, positive for the *bla_{TEM}* gene; lanes 9 to 10, positive for the *bla_{CTX-M-1}* gene.



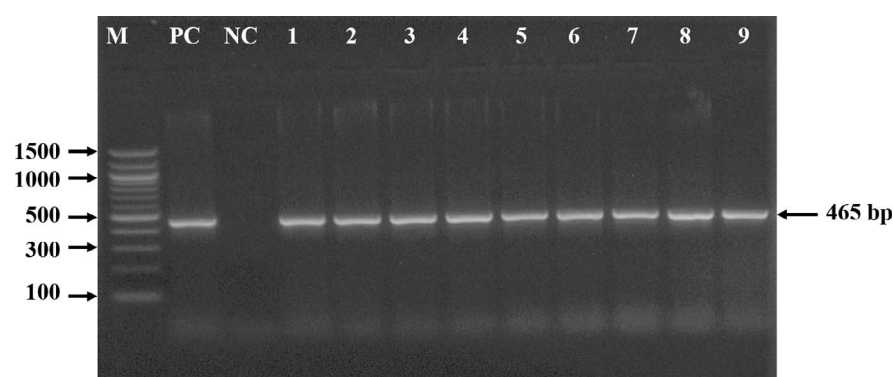
and the existence of possibly extensively drug-resistant *Salmonella* found in this study could be due to the unrestricted use of higher-end antibiotics such as colistin in poultry production for therapy, prophylaxis, and growth promotion. There are no strict laws to control the extra-label use of antibiotics on poultry farms in Bangladesh. Continuous use of antibiotics on poultry farms creates selection pressure and speeds up the emergence of MDR pathogens.

In the current study, resistance to trimethoprim-sulfamethoxazole and amoxicillin was the most common, which is consistent with previous reports (18, 24). However, the resistance percentages were higher than those reported in Nepal (55) and China (73). Tetracycline resistance was also commonly observed among the *Salmonella* isolates, and it has been reported from several other studies on frozen chicken meat (2, 68, 70). This prevalence was higher than the findings of earlier studies in Nepal and Korea (27, 55). Tetracycline has been one of the most commonly used antibiotics for poultry production, which probably explains the very frequent occurrence of resistance to tetracycline among *Salmonella*. Moreover, a high resistance rate to nalidixic acid was observed in *Salmonella* isolates, which is consistent with earlier reports (2, 24, 50). The high level of nalidixic acid resistance in this study is of special concern because it may limit the therapeutic use of fluoroquinolones. Nonsusceptibility to pefloxacin was also high in the present study. According to the recommendation by CLSI, resistance to one drug belonging to the fluoroquinolones class implies that the bacteria are resistant to the whole class (14). This information carries great value because fluoroquinolones are considered to be the drug of first choice for the treatment of *Salmonella* infections in humans (73). On the other hand, a relatively low resistance rate to cefotaxime, ceftazidime, and ceftriaxone was observed in this study,

which may be due to the fact that third-generation cephalosporins are not used in poultry production in Bangladesh, resulting in a lack of selective pressure by this antimicrobial in poultry production. Surprisingly, resistance to carbapenems (the last line of drugs to treat MDR bacteria), mainly to imipenem, was observed in 48.6% of *Salmonella* isolates; this shows that *Salmonella* has a tendency toward drug resistance because this drug is not used in poultry practices in Bangladesh. The resistance to carbapenems might be due to coresistance in other antimicrobial classes (37). The emergence of coresistance can occur either alone or in combination as a result of long-term and extensive use of broad-spectrum antimicrobial agents (25, 43). Hence, there is a need to strengthen epidemiological investigations of antimicrobial-resistant *Salmonella* infections, and further studies are required to characterize the resistance mechanisms and spread of resistant bacteria.

The β -lactamase activity in gram-negative bacteria is the major cause of resistance to β -lactam antibiotics (23). In general, β -lactamases in the *Enterobacteriaceae* are primarily TEM and CTX-M enzymes, which confer resistance to β -lactam penicillins and cephalosporins (48, 74). *bla_{TEM}* was the most commonly detected BSBL-encoding gene in the present study. Earlier reports also documented the highest occurrence of *bla_{TEM}*, followed by *bla_{CTX-M}*, in *Salmonella* isolated from retail chicken meat in China (50, 71). A report indicated that the TEM gene has been disseminating rapidly among populations of gram-negative bacteria in humans in Bangladesh (4). In addition, we found only two isolates of *Salmonella* that harbored the *bla_{CTX-M-1}* ESBL-encoding gene. This ESBL-encoding gene has been detected previously in two human *Salmonella* species isolates in Bangladesh (47). Notably, approximately 20% of *Salmonella* isolates in our study harbored carbapene-

FIGURE 3. *Carbapenemase-encoding gene (bla_{NDM-1}) of Salmonella isolates from frozen chicken meat detected by uniplex PCR.* M, 100-bp DNA ladder; PC, positive control; NC, negative control; lanes 1 to 9, positive for the *bla_{NDM-1}* gene.



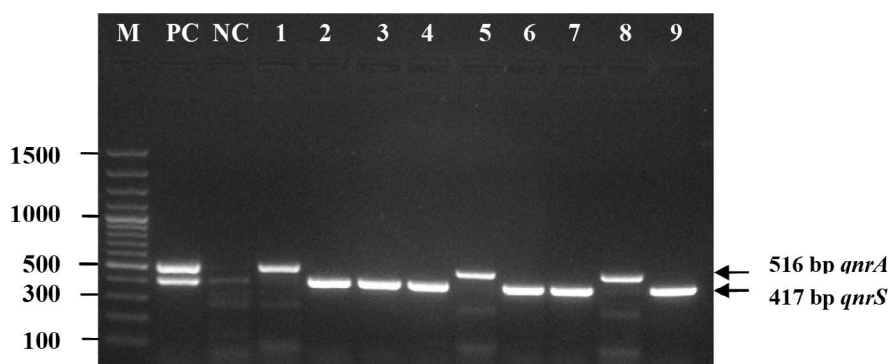


FIGURE 4. *PMQR*-encoding genes of *Salmonella* isolates from frozen chicken meat detected by multiplex PCR. M, 100-bp DNA ladder; PC, positive control (both *qnrA* and *qnrS* genes); NC, negative control; lanes 1, 5, and 8, *qnrA* gene positive (516 bp); lanes 2 to 4, 6 to 7, and 9, *qnrS* gene positive (417 bp).

mase-encoding *bla*_{NDM-1} gene, which is the first ever reported in chicken meat in Bangladesh. Prevalence of *bla*_{NDM-1} gene (29%) was reported previously in wastewater *E. coli* isolates in Bangladesh (30). The presence of *bla*_{NDM-1} is generally associated with resistance to other antimicrobial compounds, such as aminoglycosides, β -lactams, and fluoroquinolones (67).

It has been previously reported that the PMQR *qnr* genes that confer resistance to nalidixic acid and fluoroquinolones such as ciprofloxacin may impair fluoroquinolones therapy (3, 11). There are three major *qnr* genes (*qnrA*, *qnrB*, and *qnrS*) in *Enterobacteriaceae* (3, 11, 38). Among the 74 *Salmonella* isolates, *qnrA* and *qnrS* genes were detected in three and five isolates, respectively. This finding is in agreement with a similar study in Uganda, which reported that the *qnrS* gene was the most prevalent in nontyphoidal *Salmonella* isolated from the feces of chickens (44). The findings of the current study may suggest that nalidixic acid resistance in *Salmonella* isolates may not be due to these known PMQR genes, but may be associated with other unknown genes or mechanisms. Bacterial DNA gyrase (*gyrA* and *gyrB*) and topoisomerase IV (*parC* and *parE*) genes are the major targets of quinolone antimicrobial agents. Therefore, mutations in these genes are also associated with fluoroquinolone resistance in most bacterial species, including *Salmonella* (39, 72).

The use of antibiotics on chicken farms has not been well controlled in Bangladesh, and many antibiotics have been used as feed additives to increase broiler production and prevent diseases. The use of antibiotics has increased on the farms over time, which may explain the greater antibiotic resistance observed in the isolates. Among the 74 *Salmonella* isolates, not only those resistant to third- and fourth-generation cephalosporin but also those resistant to the other β -lactam antibiotics harbored the ESBL- and ESBL-encoding genes. Similarly, the occurrence of the carbapenemase-encoding gene (*bla*_{NDM-1}) was observed not only in *Salmonella* isolates resistant to carbapenem antibiotics but also in isolates resistant to cephalosporins, cephamycins, penicillins, penicillins- β -lactamase inhibitors, and antipseudomonal penicillins- β -lactamase inhibitors. Furthermore, the majority of isolates resistant to nalidixic acid harbored the *qnrA* gene. In addition, isolates resistant to other fluoroquinolone antimicrobials also contained either *qnrA* or *qnrS* or both genes. On the other hand, certain percentages of the *Salmonella* isolates that harbored the *bla*_{CTX-M-1} gene and *qnrA* or *qnrS* or both genes were not resistant against third-

and fourth-generation cephalosporins and fluoroquinolones, respectively. The discrepancy in the percentage of phenotypic and genotypic ESBL- and PMQR-encoding genes might be due to coexpression of more than one ESBL-encoding and fluoroquinolone gene in an organism (22). Some other known *bla*_{CTX-M} genes (*bla*_{CTX-M-8}, *bla*_{CTX-M-9}, and *bla*_{CTX-M-25}) and *qnr* genes (*qnrC* and *qnrD*) were not screened in this study. The present study indicates that *bla*_{TEM}, *bla*_{CTX-M-1}, *bla*_{NDM-1}, *qnrA*, and *qnrS* genes are widely distributed in *Salmonella* isolated from frozen chicken meat in Bangladesh. This might be due to the use of a single antibiotic because treatment can also select resistance to other antibiotics whose genes reside in the same mobile genetic element ("linked resistance") (1).

It would be worthwhile to sample more outlets of different supershops. However, this study sampled frozen chicken meat samples from most of the renowned supershops located in five megacities of Bangladesh; thus, the data are representative of the whole of Bangladesh. Moreover, some supershops were not available in some megacities and some outlets only sold grocery items other than chicken meat. Therefore, the collection of samples of brands 4 to 9 was not possible in some cities. The results of prevalence of *Salmonella* bacteria and their AMR pattern should be used with caution because few samples were collected and screened at a time point.

In conclusion, the data presented in this study reveal that contamination with *Salmonella* carrying ESBL was high in frozen chicken meat sampled in Bangladesh. Furthermore, all the *Salmonella* isolates were MDR. The study also reports, for the first time, the presence of the carbapenemase-encoding resistance gene (*bla*_{NDM-1}) in chicken meat in Bangladesh. Given the high rate of MDR and ESBL-producing *Salmonella* in frozen chicken meat, strict regulations and standards for chicken production and processing should be implemented before these chicken meat products are circulated in the marketplace, to ensure food safety and protect consumers' health. It is also important to strengthen the monitoring of the development and dissemination of antimicrobial-resistant *Salmonella* in animal-derived foods.

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TABLE 7. Association between antimicrobial resistance phenotypes and antimicrobial resistance genes (*-lactamase- and PMQR-encoding genes*) of *Salmonella* isolated from frozen chicken meat^a

		No. (%) of resistance genes				
		BSBL	ESBL	Carbapenemase	PMQR	
Phenotype of antimicrobial resistance	Resistant types (<i>n</i>)	<i>bla</i> _{TEM}	<i>bla</i> _{CTX-M-1}	<i>bla</i> _{NDM-1}	<i>qnrA</i>	<i>qnrS</i>
Cephalosporins						
Cephalexin	Resistant (7)	7 (100)	1 (14.3)	2 (28.5)		
	Nonresistant (67)	67 (100)	1 (1.5)	13 (19.4)		
Cephradine	Resistant (9)	9 (100)	2 (22.2)	4 (44.4)		
	Nonresistant (65)	65 (100)	0	11 (16.9)		
Cefuroxime	Resistant (2)	2 (100)	0	0		
	Nonresistant (72)	72 (100)	2 (2.8)	15 (20.8)		
Cefaclor	Resistant (10)	10 (100)	1 (10.0)	3 (30.0)		
	Nonresistant (64)	64 (100)	1 (1.6)	12 (18.8)		
Cefotaxime	Resistant (3)	3 (100)	1 (33.3)	1 (33.3)		
	Nonresistant (71)	71 (100)	1 (1.4)	14 (19.7)		
Ceftriaxone	Resistant (2)	2 (100)	1 (50.0)	1 (50.0)		
	Nonresistant (72)	72 (100)	1 (1.4)	14 (19.4)		
Ceftazidime	Resistant (1)	1 (100)	0	0		
	Nonresistant (73)	73 (100)	2 (2.7)	15 (20.5)		
Cefixime	Resistant (4)	4 (100)	0	2 (50.0)		
	Nonresistant (70)	70 (100)	2 (2.9)	13 (18.6)		
Cefepime	Resistant (1)	1 (100)	0	0		
	Nonresistant (73)	73 (100)	2 (2.7)	15 (20.5)		
Cephameycins						
Cefoxitin	Resistant (6)	6 (100)	1 (16.7)	2 (33.3)		
	Nonresistant (68)	68 (100)	1 (1.5)	13 (19.1)		
Penicillins						
Ampicillin	Resistant (47)	47 (100)	2 (4.3)	13 (27.7)		
	Nonresistant (27)	27 (100)	0	2 (7.4)		
Amoxycillin	Resistant (55)	55 (100)	2 (3.6)	14 (25.5)		
	Nonresistant (19)	19 (100)	0	1 (5.3)		
Penicillins + β-lactamase inhibitors						
Amoxicillin–clavulanic acid	Resistant (19)	19 (100)	0	7 (36.8)		
	Nonresistant (55)	55 (100)	2 (3.6)	8 (14.5)		
Antipseudomonal penicillins + β-lactamase inhibitors						
Piperacillin-tazobactam	Resistant (15)	15 (100)	1 (6.7)	5 (33.3)		
	Nonresistant (59)	59 (100)	1 (1.7)	10 (16.9)		
Carbapenems						
Imipenem	Resistant (36)	36 (100)	2 (5.6)	13 (36.1)		
	Nonresistant (38)	38 (100)	0	2 (5.3)		
Meropenem	Resistant (10)	10 (100)	0	3 (30.0)		
	Nonresistant (64)	64 (100)	2 (3.1)	12 (18.8)		
Fluoroquinolones						
Nalidixic acid	Resistant (62)				1 (1.6)	3 (4.8)
	Nonresistant (12)				2 (16.7)	2 (16.7)
Ciprofloxacin	Resistant (28)				0	1 (3.6)
	Nonresistant (46)				3 (6.5)	4 (8.7)
Levofloxacin	Resistant (8)				0	1 (12.5)
	Nonresistant (66)				3 (4.5)	4 (6.1)
Norfloxacin	Resistant (14)				0	1 (7.1)
	Nonresistant (60)				3 (5.0)	4 (6.7)
Ofloxacin	Resistant (26)				1 (3.8)	1 (3.8)
	Nonresistant (48)				2 (4.2)	4 (8.3)
Gatifloxacin	Resistant (22)				1 (4.5)	1 (4.5)
	Nonresistant (52)				2 (3.8)	4 (7.7)
Pefloxacin	Resistant (52)				2 (3.8)	2 (3.8)
	Nonresistant (22)				1 (4.5)	3 (13.6)

^a n, number of isolates; BSBL, broad-spectrum β -lactamase; ESBL, extended-spectrum β -lactamase; *bla*_{NDM}, New Delhi metallo- β -lactamase; PMQR, plasmid-mediated quinolone resistance.

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